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Memory-Augmented Scene Understanding and Exploration for Open-World Aerial Object-Goal Navigation

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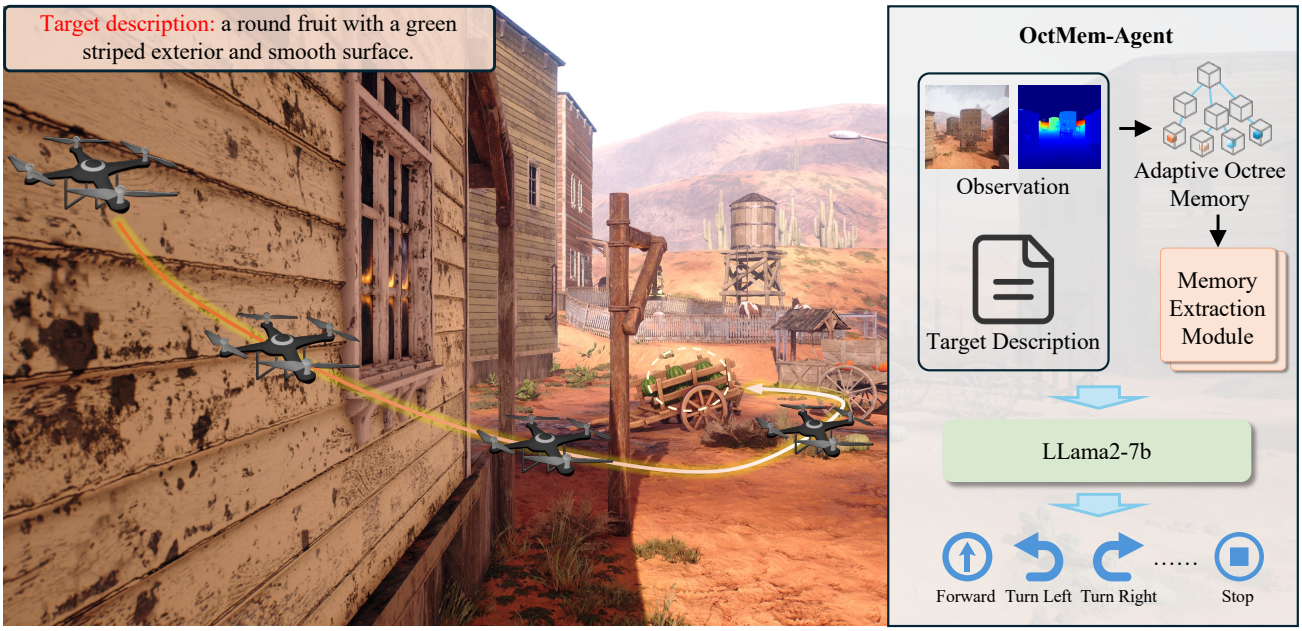


Figure 1. Overview of our work. As shown on the left, aerial object-goal navigation requires an Unmanned Aerial Vehicle (UAV) to navigate to target objects in large-scale outdoor environments using only visual observations and high-level object descriptions. In this work, we propose OctMem-Agent, an octree memory-augmented framework illustrated on the right. OctMem-Agent constructs an Adaptive Octree Memory from RGB-D observations and employs a Memory Extraction Module to extract compact information that facilitates navigation decision-making.

Abstract

Aerial object-goal navigation (Aerial ObjectNav) requires an Unmanned Aerial Vehicle (UAV) to navigate to target objects in large-scale outdoor environments using only visual observations and high-level object descriptions, without detailed step-by-step instructions. Existing approaches rely on local observations or short-term history, lacking comprehensive scene understanding and effective spatial exploration strategies, which constrains their navigation capability in complex aerial scenarios. To address these challenges, we propose ***OctMem-Agent****, an octree memory-augmented framework for Aerial ObjectNav. Specifically, we introduce an Adaptive Octree Memory Representation that incrementally aggregates RGB-D observations into a hierarchical 3D representation, capturing both explored regions and unexplored frontiers across large-scale aerial environments. We further propose a Instruction-Guided Memory Query module that extracts task-relevant scene and exploration tokens through instruction-modulated queries. By integrating these tokens with visual observations and language instructions, OctMem-Agent achieves comprehensive*

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scene understanding and effective spatial exploration for target localization. Extensive experiments on the Aerial ObjectNav benchmark UAV-ON demonstrate that our method achieves a significant 7.5% improvement in success rate over existing methods, validating the effectiveness of our design.

1. Introduction

Building on recent advances in aerial Vision-and-Language Navigation (VLN) [7, 8], Aerial object-goal navigation (Aerial ObjectNav) enables UAVs to navigate and locate target objects based on visual observations and high-level object descriptions [41], rather than detailed, step-by-step instructions. This capability is critical for deploying UAVs in emergency response, search and rescue, and package delivery scenarios [2, 9, 56]. Such applications often feature rapidly changing environmental conditions, unavailable communication infrastructure, and impractical human supervision, requiring UAVs to autonomously locate target objects. However, achieving robust Aerial ObjectNav poses unique challenges. Compared to ground-based ObjectNav [23, 40] in indoor settings, Aerial ObjectNav operates in large-scale outdoor environments characterized by complex terrain and dramatic viewpoint variations [7], which demands scale-adaptive scene perception and effective spatial exploration strategies to effectively locate targets.

Existing methods [8, 41] rely on local observations and short-term history, lacking global scene understanding and long-term spatial memory, thus hindering effective navigation in complex aerial scenarios. For instance, AOA [41] generates navigation actions via LLMs using current observations and recent pose history, while OpenFly [8] operates with only the current frame and the previous two frames. This often results in myopic decisions and redundant exploration across large-scale outdoor environments. Furthermore, scene representations developed for ground-based indoor navigation [18, 30, 40] (e.g., volumetric maps [24], semantic maps [23, 36]) cannot be directly applied to aerial scenarios due to drastically larger spatial scales and altitude-dependent viewpoint variations [53].

To address these challenges, we propose OctMem-Agent, an octree memory-augmented vision-language navigation framework for aerial object-goal navigation. Specifically, we introduce an Adaptive Octree Memory Representation that incrementally aggregates RGB-D observations into a hierarchical 3D representation. The memory construction consists of three stages: first projecting 2D visual features into 3D patches with geometric information [57], then hierarchically aggregating patches using distance-adaptive voxelization that balances fine-grained local details with compact distant representations, and finally integrating the aggregated features into an octree-based memory structure. This hierarchical representation enables effective spatial indexing and scalable memory management across large-scale aerial environments, providing comprehensive scene understanding for long-term navigation.

Building upon this memory representation, we introduce an Instruction-Guided Memory Query module that extracts task-relevant information through instruction-modulated queries. The module initializes two types of learnable queries—scene queries and exploration queries—and employs Feature-wise Linear Modulation (FiLM) [28] to condition them on language instructions. These modulated queries then attend to different regions of the memory via a Q-Former [21]: scene queries focus on nearby memory voxels for target localization, while exploration queries attend to distant voxels for identifying unexplored frontiers. By integrating the extracted memory representations with current visual observations and language instructions, OctMem-Agent achieves comprehensive scene understanding and effective spatial exploration. We conduct extensive experiments on the challenging Aerial ObjectNav benchmark UAV-ON [41]. Results demonstrate that OctMem-Agent achieves a significant 7.5% improvement in success rate over existing methods, validating the effectiveness of our memory-augmented approach. Overall, the main contributions of our work are as follows:

- We present an Adaptive Octree Memory that incrementally aggregates RGB-D observations into hierarchical 3D representations for scene understanding in large-scale environments.
- We introduce an Instruction-Guided Memory Query module that extracts complementary scene and exploration representations through instruction-modulated queries, effectively balancing target localization and frontier exploration.
- Extensive experiments on Aerial ObjectNav benchmark UAV-ON show that our method achieves state-of-the-art performance with a significant improvement in success rate.

2. Related Works

Aerial Object-goal Navigation. Traditional object-goal navigation (ObjectNav) [1] requires an agent to navigate towards a specified object category based on visual observations and language instructions [45]. Early works primarily focused on indoor environments such as AI2-THOR [17], Gibson [40], RoboTHOR [6], and HM3D [30]. Methodologically, these approaches evolved from end-to-end reinforcement learning and imitation learning [31, 43, 44] to methods leveraging multimodal large language models [22, 33, 52] for semantic understanding and spatial reasoning [37, 47, 48]. More recently, zero-shot ObjectNav

methods leverage Vision-Language Models (VLMs) to generalize towards unseen object categories [4, 54, 55].

Recently, aerial object-goal navigation (Aerial ObjectNav) has emerged as a new frontier [19, 41, 51], presenting unique challenges compared to ground-based navigation. Unlike constrained indoor environments, Aerial ObjectNav operates in large-scale outdoor scenes with complex terrain and dynamic environmental changes, demanding sophisticated spatial reasoning across unrestricted three-dimensional space [8, 25, 35]. Current approaches to aerial object-goal navigation, such as AOA [41],GeoNav [42], primarily focus on zero-shot generalization by leveraging Vision-Language Models (VLMs) to handle diverse high-level goals in unconstrained outdoor environments. In contrast, our method builds upon pretrained VLMs and integrates an Adaptive Octree Memory Representation for scene representation and spatial exploration in complex aerial environments.

Memory-based Vision Language Navigation. Spatial memory plays a crucial role in vision-language navigation by enabling agents to build and maintain scene representations [20]. Existing spatial memory methods can be broadly categorized into dense representations and sparse representations. Dense representations include early semantic maps [3, 12] built with task-specific networks for limited object categories, as well as more recent voxel maps [24] and neural feature fields [26, 32] leveraging pretrained vision-language models [13, 29, 50]. These methods provide fine-grained spatial encoding but incur substantial memory overhead. Sparse representations, such as topological graphs [23], grid maps [5, 36], and hierarchical graph structures [10, 38, 39], offer more compact scene encoding through structured abstractions.

However, both paradigms face significant challenges when scaling to open-world aerial 3D navigation. Dense representations struggle with large spatial extents, as their memory requirements grow cubically with scene size, making them impractical for outdoor environments with dynamic conditions and diverse terrain. Sparse graph-based representations are limited by their design assumptions. Their discrete spatial abstractions and hierarchical structures, optimized for indoor object-level scene understanding, do not generalize well to large-scale outdoor scenarios.

3. Method

In this section, we present **OctMem-Agent**, a vision-language navigation framework for aerial object-goal navigation. As Figure 2 shows, the framework comprises three key components: (1) Adaptive Octree Memory Representation that aggregates historical RGB-D observations into a scalable 3D representation, (2) Instruction-Guided Memory Query that extracts compact scene and exploration tokens via a FiLM-modulated Q-Former, and (3) Action Generation with Memory Integration that combines memory, observation, and language instruction to predict actions.

3.1. Problem Formulation

Given a language instruction $I_{\text{goal}} = \{w_1, \ldots, w_{N_I}\}$ consisting of N_I words that describes a navigation objective, Aerial ObjectNav aims to guide an aerial agent from an initial 3D pose $p_0 = (x_0, y_0, z_0, \phi_0)$ to the target location by executing a sequence of actions. Here, $(x_0, y_0, z_0) \in \mathbb{R}^3$ denotes the initial position and ϕ_0 represents the yaw angle. At each time step t, the agent receives an observation $O_t = \{\mathbf{D}_t, \mathbf{V}_t\}$, where $\mathbf{D}_t$ and $\mathbf{V}_t$ denote the depth map and RGB image respectively, and executes an action $a_t \in \mathcal{A}$ conditioned on I_{goal} and O_t to update its pose to p_{t+1}. Following [8, 41], we define the discrete action space as $\mathcal{A} = \{\texttt{Move Forward}, \texttt{Turn Left}, \texttt{Turn Right}, \texttt{Ascend}, \texttt{Descend}, \texttt{Stop}\}$, where forward movement(*e.g.*, `Move Forward`) translates the agent by 3 units, vertical movement(*e.g.*, `Ascend`) adjusts altitude by 3 meters, and turning rotates the agent by 30 degrees. The episode terminates when the agent executes a `Stop` action or reaches the maximum step limit. Consistent with previous work [8, 41, 53], the navigation is considered successful if the agent is within 20 meters of the destination.

3.2. Adaptive Octree Memory Representation

To achieve effective long-term spatial representation and reasoning in large-scale 3D aerial environments, we propose an **Adaptive Octree Memory** that aggregates historical observations into a unified and hierarchical memory representation. Our memory construction process consists of three stages: (1) 3D Patches Representation, (2) Hierarchical Spatial Aggregation, and (3) Memory Integration.

3D Patches Representation. To enable better 3D understanding, we construct 3D patches by injecting 3D positional information into 2D patches. This allows the 3D patches to explicitly capture spatial structure while retaining the semantic information from the original 2D patches. Specifically, for each incoming observation $O_t = \{\mathbf{D}_t, \mathbf{V}_t\}$ at timestep t, a vision encoder extracts patch token features from the RGB image $\mathbf{V}_t$ and reshapes them into a spatial grid $\mathbf{X}_p \in \mathbb{R}^{h \times w \times d}$. Here $h \times w$ is the spatial resolution and d is the feature dimension. Following [14, 57], we back-project these patches to obtain their 3D world coordinates $\mathbf{P} \in \mathbb{R}^{h \times w \times 3}$ using the camera's intrinsic and extrinsic parameters and nearest neighbor depth values from $\mathbf{D}_t$. The 3D positions $\mathbf{P}$ are then encoded into 3D positional embeddings $\mathbf{P}' \in \mathbb{R}^{h \times w \times d}$ via a learnable two-layer MLP. We obtain the 3D patches by fusing the positional embeddings with the 2D features:

$$\mathbf{X}_{3D} = \mathbf{X}_p + \mathbf{P}'. \tag{1}$$

Hierarchical Spatial Aggregation. The raw 3D patches

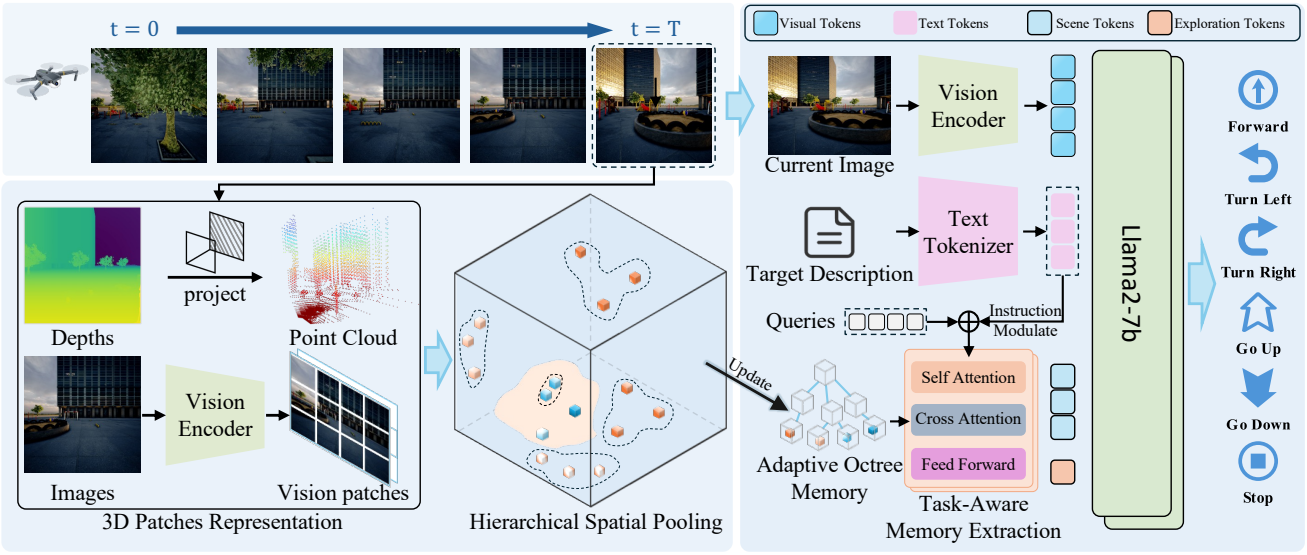


Figure 2. The overall framework of our proposed OctMem-Agent. The agent builds an Adaptive Octree Memory from historical RGB-D data. The instruction-guided query module extracts compact scene and exploration tokens. These tokens are fused with the current visual observation and language instruction, serving as input to the LLM for action inference.

are often overly dense, making direct processing computationally prohibitive. We employ hierarchical pooling to adaptively aggregate spatially important patches.

In Aerial ObjectNav tasks, points beyond the agent's effective planning range are spatially irrelevant as they provide limited navigational value (e.g., points beyond 10000 meters are unlikely to be useful when searching for objects within a 100-meter range). We thus filter out points excessively far from the current agent position (x_t, y_t, z_t). Specifically, we remove points from $\mathbf{P}$ whose distance to (x_t, y_t, z_t) exceeds a threshold $d_{\max}$, and apply the same filtering to $\mathbf{X}_{3D}$ accordingly, obtaining the filtered sets $\hat{\mathbf{P}}$ and $\hat{\mathbf{X}}_{3D}$. Since precise navigation requires detailed local information while effective exploration demands compact representation of distant space, we represent nearby regions with finer-grained voxels to preserve geometric and semantic details, and distant regions with coarser voxels to capture large-scale layout and unexplored frontiers. To achieve this, we partition the filtered points $\hat{\mathbf{P}}$ into hierarchical voxel grids with varying sizes determined by their distance from the agent. Specifically, we define distance-based intervals:

$$\mathcal{D} = \{[d_0, d_1), [d_1, d_2), \ldots, [d_{K-1}, d_K]\}, \tag{2}$$

where $d_0 = 0$ and $d_K = d_{\max}$, and K is the number of intervals. Each interval $[d_{k-1}, d_k)$ corresponds to a voxel size s_k with $s_1 < s_2 < \cdots < s_K$. Within each voxel v, both the 3D points and their associated patch features are aggregated via mean pooling:

$$\mathbf{x}_v = \frac{1}{|\mathcal{I}_v|} \sum_{i \in \mathcal{I}_v} \mathbf{x}_i, \quad \mathbf{p}_v = \frac{1}{|\mathcal{I}_v|} \sum_{i \in \mathcal{I}_v} \mathbf{p}_i, \tag{3}$$

where $\mathcal{I}_v$ denotes the set of indices in voxel v. This yields the final aggregated representations $\mathbf{X}_{\text{agg}} = \{\mathbf{x}_v | v \in \mathcal{V}\}$ and $\mathbf{P}_{\text{agg}} = \{\mathbf{p}_v | v \in \mathcal{V}\}$, where $\mathcal{V}$ is the set of all voxels.

Memory Integration. The aggregated patches $\mathbf{X}_{\text{agg}}$ and points $\mathbf{P}_{\text{agg}}$ are incrementally integrated into a global octree-based memory structure $\mathcal{M}_t$. The octree structure enables effective spatial indexing and hierarchical representation of the 3D environment. Each octree cell corresponds to a spatial region and stores the aggregated patch features of points within that region, allowing fast retrieval and update of local representations.

Initially at $t = 0$, we construct the octree memory from the first set of aggregated points $\mathbf{P}_{\text{agg}}^{(0)}$ by recursively partitioning the 3D space into octants based on spatial occupancy. Each point i is assigned to a cell c, with the cell feature initialized as $\mathbf{m}_c^{(0)} = \mathbf{x}_i^{(0)}$, forming the initial memory $\mathcal{M}_0 = \{\mathbf{m}_c^{(0)} | c \in \mathcal{C}_0\}$. At subsequent timesteps $t > 0$, new points $\mathbf{P}_{\text{agg}}^{(t)}$ and features $\mathbf{X}_{\text{agg}}^{(t)}$ are inserted into the octree by matching them to their corresponding spatial cells based on 3D coordinates. For points matched to existing cells, the corresponding cell features are updated by averaging with their associated features from $\mathbf{X}_{\text{agg}}^{(t)}$. For each point j in previously unvisited regions, a new octree cell c' is created and initialized as $\mathbf{m}_{c'}^{(t)} = \mathbf{x}_j^{(t)}$. This yields the updated memory representation $\mathcal{M}_t = \{\mathbf{m}_c^{(t)} | c \in \mathcal{C}_t\}$, where $\mathcal{C}_t$ denotes the set of all cells (both existing and newly created) at timestep t.

3.3. Instruction-Guided Memory Query

In this section, we present a method to extract task-relevant information from $\mathcal{M}_t$ based on instruction I_{goal} via two

key components: (1) Instruction-Modulated Query and (2) Task-Aware Memory Extraction.

Instruction-Modulated Query. We initialize a set of learnable query embeddings $\mathbf{Q} \in \mathbb{R}^{N_q \times d}$, where N_q is the number of queries and d matches the dimension of visual patch embeddings.

To enable task-specific memory querying, we employ Feature-wise Linear Modulation (FiLM) [16, 28] to inject instruction information into the queries. Specifically, we encode the natural language instruction I_{goal} into an embedding using a pre-trained language encoder, followed by average pooling to obtain a d-dimensional vector $\mathbf{e}_I \in \mathbb{R}^d$. Then, we compute feature-wise scaling and shifting parameters $\boldsymbol{\gamma}(\mathbf{e}_I) \in \mathbb{R}^d$ and $\boldsymbol{\beta}(\mathbf{e}_I) \in \mathbb{R}^d$ via linear projections.

These parameters modulate each dimension of the initial queries $\mathbf{Q} \in \mathbb{R}^{N_q \times d}$ through an affine transformation:

$$\mathbf{Q}_{\text{task}} = \text{FiLM}(\mathbf{Q}, \mathbf{e}_I) = (1 + \boldsymbol{\gamma}(\mathbf{e}_I)) \odot \mathbf{Q} + \boldsymbol{\beta}(\mathbf{e}_I), \quad (4)$$

where $\odot$ denotes element-wise multiplication along each feature dimension. This conditioning ensures that the subsequent memory attention operates with respect to the navigation goal.

Task-Aware Memory Extraction. The queries $\mathbf{Q}_{\text{task}}$ are then used to extract relevant information from the global memory $\mathcal{M}_t$ via a Q-Former [21].

To effectively balance target localization and frontier exploration, $\mathbf{Q}_{\text{task}}$ consists of two distinct subsets, each serving complementary purposes: (1) **Scene tokens** $\mathbf{Q}_{\text{scene}} \in \mathbb{R}^{N_s \times d}$, which attend to memory voxels within a specified boundary range, focusing on semantic objects that match the target description in I_{goal} to enable precise target localization. (2) **Exploration tokens** $\mathbf{Q}_{\text{explore}} \in \mathbb{R}^{N_e \times d}$, which attend to memory voxels beyond this boundary, identifying frontiers between explored and unexplored regions to guide the agent towards potential search areas when the target is not immediately visible. Here, $N_s + N_e = N_q$, and both subsets undergo the same FiLM modulation described in Eq. 4. We select a distance threshold d_b from the interval set $\mathcal{D}$ (Eq. 2) to define this boundary range, where scene tokens query voxels within distance d_b from the agent, while exploration tokens query voxels beyond d_b.

Within the Q-Former, queries are processed through L Transformer layers, each consisting of self-attention followed by cross-attention with the memory. During cross-attention, we partition the memory $\mathcal{M}_t$ into two subsets:

$$\begin{aligned} \mathcal{M}_{\text{near}} &= \{\mathbf{m}_c^{(t)} \in \mathcal{M}_t \mid \text{dis}(c) < d_b\}, \\ \mathcal{M}_{\text{far}} &= \{\mathbf{m}_c^{(t)} \in \mathcal{M}_t \mid \text{dis}(c) \geq d_b\}, \end{aligned} \quad (5)$$

where $\text{dis}(c)$ denotes the Euclidean distance from the agent to cell c, and d_b is the boundary distance threshold. Scene tokens $\mathbf{Q}_{\text{scene}}$ perform cross-attention exclusively with $\mathcal{M}_{\text{near}}$ to capture local spatial context, while exploration tokens $\mathbf{Q}_{\text{explore}}$ attend only to $\mathcal{M}_{\text{far}}$ to identify distant unexplored regions. This yields complementary representations $\mathbf{Q}_{\text{scene}}^{(L)}$ and $\mathbf{Q}_{\text{explore}}^{(L)}$ that guide subsequent navigation decisions.

3.4. Action Generation with Memory Integration

Our OctMem-Agent builds upon the OpenVLA [15] framework, adapting it for discrete action prediction in aerial object-goal navigation. To integrate the extracted memory into the model, the Q-Former outputs $\mathbf{Q}_{\text{scene}}^{(L)}$ and $\mathbf{Q}_{\text{explore}}^{(L)}$ are concatenated and projected through a linear layer to match the VLM's hidden dimension. This yields memory representations $\mathbf{H}_{\text{mem}}$. To incorporate the current observation, we encode the RGB image $\mathbf{V}_t$ using the same vision encoder as in Section 3.2 to obtain 2D patch features $\mathbf{H}_{\text{obs}}$. The language instruction I_{goal} is processed through a pre-trained language encoder to produce text embeddings $\mathbf{H}_{\text{lang}}$. These multimodal representations are concatenated along the sequence dimension to form a unified input sequence $\mathbf{H}_{\text{input}} = [\mathbf{H}_{\text{lang}}, \mathbf{H}_{\text{obs}}, \mathbf{H}_{\text{mem}}]$. Following OpenVLA [15], we feed $\mathbf{H}_{\text{input}}$ into a pre-trained vision-language model (VLM) backbone, which processes the sequence through its transformer layers to generate predicted action tokens. Finally, an action decoder discretizes the predicted tokens into discrete actions $a_t \in \mathcal{A}$.

4. Experiments

In this section, we conduct comprehensive experiments on aerial object navigation benchmarks to evaluate the effectiveness of our OctMem-Agent. Extensive ablation studies are also conducted to show the effectiveness of our Adaptive Octree Memory Representation and the Instruction-Guided Memory Query.

4.1. Experimental Settings

Benchmark. We primarily evaluate our method on UAV-ON [41], a comprehensive benchmark designed for aerial object-goal navigation in complex and diverse environments. The benchmark contains 14 high-fidelity outdoor environments developed in Unreal Engine, spanning diverse geographic structures from compact urban parks to expansive mountainous terrains, including villages, towns, cities, forests, and water areas. It contains 1,270 target objects with corresponding high-level instructions, with the objects varying in size from small to large.

The benchmark comprises 10,000 training episodes across 10 environments and 1,000 test episodes spanning the 10 training environments and 4 additional held-out environments. The test split includes novel objects and diverse instructions to assess generalization capabilities. Each episode consists of a random starting position and a goal object described by a natural language instruction.

Table 1. Performance comparison on the UAV-ON benchmark across different target object sizes. Our OctMem-Agent achieves the highest success rates across all object sizes and surpasses all baselines in total performance.

Method	Small			Medium			Large			Total		
	SR↑	OSR↑	SPL↑	SR↑	OSR↑	SPL↑	SR↑	OSR↑	SPL↑	SR↑	OSR↑	SPL↑
Random	4.14%	7.80%	2.80%	3.33%	8.10%	3.05%	2.48%	8.07%	1.62%	3.70%	8.00%	2.66%
CLIP-H [41]	2.86%	8.43%	1.51%	10.95%	16.67%	**7.17%**	13.04%	19.25%	10.53%	6.20%	11.90%	4.15%
AOA-F [41]	4.45%	16.38%	1.61%	10.48%	17.62%	6.36%	14.29%	21.74%	10.66%	7.30%	17.50%	4.06%
AerialVLN [7]	4.13%	19.87%	4.01%	4.76%	17.62%	4.9%	1.86%	19.25%	1.86%	3.9%	19.30%	3.72%
Navid [49]	10.02%	29.55%	4.41%	11.43%	26.44%	5.95%	17.39%	30.73%	**15.03%**	11.50%	29.10%	**6.44%**
OpenFly [8]	12.40%	26.23%	**6.35%**	10.00%	20.48%	4.19%	13.04%	31.65%	7.56%	12.00%	25.90%	6.09%
OctMem-Agent	**18.91%**	**29.57%**	6.03%	**18.09%**	**26.66%**	6.27%	**23.60%**	**31.67%**	7.80%	**19.50%**	**29.30%**	6.37%

Table 2. Performance comparison on seen and unseen environments. Our method consistently outperforms baselines on SR and OSR across both settings.

Method	Seen			Unseen		
	SR↑	OSR↑	SPL↑	SR↑	OSR↑	SPL↑
AerialVLN	4.55%	16.84%	4.24%	3.51%	20.77%	3.41%
Navid	14.17%	30.08%	7.19%	10.70%	28.51%	**5.99%**
OpenFly	12.29%	26.20%	6.83%	11.82%	25.71%	5.64%
OctMem-Agent	**22.72%**	**30.48%**	**8.40%**	**17.57%**	**28.59%**	5.15%

Table 3. Ablation study on key components. We evaluate the impact of Adaptive Octree Memory Representation and Instruction-Guided Memory Query on model performance.

Adaptive Octree Memory Representation	Instruction-Guided Memory Query	SR↑	OSR↑	SPL↑
×	×	12.40%	20.60%	3.35%
✓	×	15.70%	21.10%	4.35%
✓	✓	**19.50%**	**29.30%**	**6.37%**

Since UAV-ON provides only annotated data without trajectory information, we generate training trajectories following the approaches described in [41], which employs a three-dimensional A* algorithm [11] to compute the shortest traversable path to each target object. We then use these trajectories to train both OctMem-Agent and baseline methods.

Baselines. We compare our approach against two categories of methods. (i) We adapt existing aerial vision-language navigation models to the Aerial ObjectNav task on UAV-ON, including aerial navigation methods (OpenFly [8], AerialVLN [7]) and a state-of-the-art indoor VLN model (Navid [49]). These methods are trained on the UAV-ON training set. (ii) We compare against zero-shot aerial object-goal navigation approaches, including CLIP-H [41] and AOA-F [41], which leverage vision-language foundation models for decision-making. Additionally, we include a random baseline that randomly selects actions at each step until the `Stop` action is chosen or the maximum step limit is reached, serving as a reference. We provide additional experimental results in the Appendix.

Metrics. We evaluate agent performance using three metrics: Success Rate (SR), Oracle Success Rate (OSR), and Success-weighted Path Length (SPL). These metrics are widely used in embodied navigation tasks [30, 47]. The detailed definitions are as follows:

- **Success Rate (SR)** measures whether the agent stops within a threshold distance from the goal object. Following [19, 41], an episode is considered successful if the final distance between the agent and the target is less than the success threshold of $\tau = 20$ meters.
- **Oracle Success Rate (OSR)** measures whether the agent has ever come within the success threshold during the episode, regardless of when it executes the Stop action. This metric captures the agent's capacity to explore and reach near the goal during its trajectory.
- **Success-weighted Path Length (SPL)** is a weighted success rate where each successful episode is weighted by the ratio of the shortest path length to the actual trajectory length, measuring both navigation success and path efficiency.

Implementation details. We build our method upon OpenVLA [15] and initialize it from the same pretrained weights as OpenFly [8] for fair comparison. Consistent with OpenVLA, our shared encoder uses a vision encoder that concatenates features from DINOv2 [27] and SigLIP [46], while adopting LLaMA-2 7B [34] as the LLM backbone. Statistical analysis of the UAV-ON benchmark reveals that distances between target objects and initial positions are consistently below 500 meters. Based on this observation, we set $d_{\max} = 500$. We set the frontier partitioning distance threshold to $d_b = 50$ and establish distance intervals $\mathcal{D} = [0, d_b), [d_b, d_{\max})$ with corresponding step sizes $s_k \in \{5, 25\}$. For the query settings, we use a total of $N_q = 144$ task queries, consisting of $N_s = 128$ scene queries and $N_e = 16$ explore queries. We train OctMem-Agent and baselines with a batch size of 64 and a learning rate of 2e-5 for 2 epochs on four L20 GPUs.

4.2. Main Results

Performance Comparison. Table 1 presents a comprehensive performance comparison of our OctMem-Agent against existing methods across different target object sizes (Small, Medium, and Large). Our OctMem-Agent demonstrates superior performance across all baseline categories and object sizes. Compared to zero-shot object-

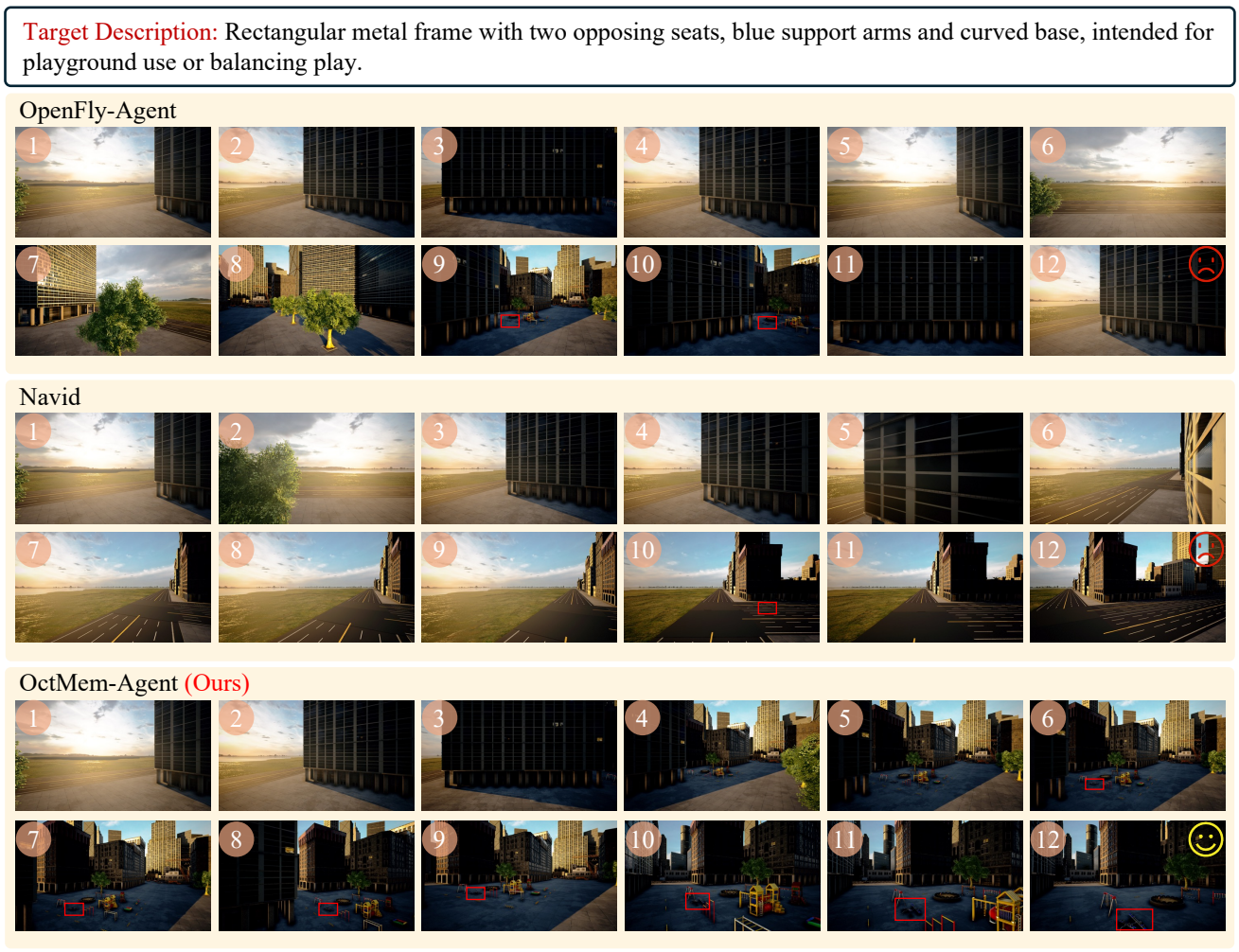


Figure 3. We present a qualitative comparison of our OctMem-Agent and baseline methods on UAV-ON. OctMem-Agent maintains consistent detection throughout the entire sequence and successfully reaches the goal, while OpenFly-Agent gets stuck rotating in place and loses the target, and Navid fails to locate the object entirely. This highlights the effectiveness of our OctMem-Agent in achieving robust tracking and successful navigation.

goal navigation approaches, our method significantly outperforms CLIP-H by 13.40% in SR and 17.40% in OSR, and surpasses AOA-F by 12.20% in SR and 11.80% in OSR on the total results. When compared to trained aerial vision-language navigation models, OctMem-Agent achieves 7.50% and 8.00% higher SR than state-of-the-art methods OpenFly [8] and Navid [49], with corresponding improvements of 3.40% and 0.20% in OSR, respectively. Notably, our method achieves the highest success rates across all object sizes (Small, Medium, and Large) and surpasses all baselines in total performance, validating the effectiveness of our framework. Our method also achieves favorable results in the SPL metric. These results validate the effectiveness of our approach in handling complex aerial object navigation tasks across diverse target object sizes.

Generalization Capability Analysis. We validate our method's generalization capability on the UAV-ON benchmark, which evaluates performance on both seen environments and unseen environments that contain novel objects and diverse instructions. As shown in Table 2, OctMem-Agent demonstrates robust generalization across multiple metrics. On seen environments, our method achieves superior performance with 22.72% SR, 30.48% OSR, and 8.4% SPL, significantly outperforming all baseline methods. More importantly, on unseen environments, OctMem-Agent maintains its advantage with 17.57% SR and 28.59% OSR, consistently surpassing other approaches. This consistent superiority in both seen and unseen settings demonstrates that our OctMem-Agent effectively captures transferable spatial knowledge and navigation strategies. The strong performance on unseen environments validates our method's ability to generalize to novel scenarios with previously unseen target objects and diverse environmental configurations.

Case Study. We present a qualitative comparison of our OctMem-Agent and baseline methods on the UAV-ON benchmark. As shown in Figure 3, our OctMem-Agent first explores the surrounding environment, then gradually narrows down potential recreational areas before precisely locating the target and successfully reaching the goal (frame 12). In contrast, OpenFly-Agent gets stuck rotating in place (frames 9-10), losing the target due to poor localization capability. Navid fails to locate the object and moves forward into an open field, revealing insufficient exploration capability. These results demonstrate our method's ability to simultaneously maintain spatial memory for target localization and exploration in large-scale complex environments.

4.3. Ablation Studies

Table 4. Ablation study on spatial aggregation methods. We compare our proposed Hierarchical Spatial Aggregation with standard voxelization baseline.

Spatial Aggregation Method	SR	OSR	SPL
Voxelization	19.10%	27.50%	5.70%
Hierarchical Spatial Aggregation	**19.50%**	**29.30%**	**6.37%**

4.3.1. Ablation Study on Main Components

To validate the effectiveness of our proposed components, we conduct an ablation study as shown in Table 3. The baseline model achieves 12.40% SR, 20.60% OSR, and 3.35% SPL. Introducing the Adaptive Octree Memory Representation yields consistent improvements across all metrics (15.70% SR, 21.10% OSR, 4.35% SPL), demonstrating that multi-scale spatial encoding captures both fine-grained details and global scene structure essential for navigation. Adding Instruction-Guided Memory Query on top of the memory representation brings additional gains, achieving 19.50% SR (+3.80%), 29.30% OSR (+8.20%), and 6.37% SPL (+2.02%). This indicates that instruction-guided queries effectively extract task-relevant information from the adaptive octree memory representation, reducing unnecessary exploration. The improvements validate the effectiveness of each component in our framework.

4.3.2. Ablation Study on Adaptive Octree Memory Representation

We compare our Hierarchical Spatial Aggregation with standard voxelization in Table 4. Standard voxelization partitions 3D space into uniform grid cells for feature aggregation. We set its voxel size to 5 to match the finest granularity in our hierarchical method. While voxelization achieves competitive performance (19.10% SR, 27.50% OSR), our hierarchical approach demonstrates notable improvements, particularly in OSR (29.30%, +1.80%) and SPL (6.37%, +0.67%). By maintaining fine-grained details for nearby regions while abstracting distant areas, the hierarchical representation enables the agent to make more informed local decisions while maintaining global scene awareness. The gains in OSR and SPL indicate that hierarchical aggregation improves the quality of navigation trajectories.

Table 5. Ablation study on Instruction-Guided Memory Query. We compare performance with and without Instruction-Modulated Query (IMQ).

Method	SR (%)	OSR (%)	SPL (%)
OctMem-Agent$_{\text{w/o IMQ}}$	18.60	27.30	5.97
OctMem-Agent	**19.50**	**29.30**	**6.37**

4.3.3. Ablation Study on Instruction-Guided Memory Query

To validate the effectiveness of instruction modulation in our memory query mechanism, we conduct an ablation study by comparing the full model with a variant that uses vanilla learnable queries without FiLM-based instruction conditioning. As shown in Table 5, removing the Instruction-Modulated Query (IMQ) leads to a performance drop across all metrics. Specifically, OctMem-Agent$_{\text{w/o IMQ}}$ achieves 18.60% SR, 27.30% OSR, and 5.97% SPL, while the full OctMem-Agent achieves 19.50% SR, 29.30% OSR, and 6.37% SPL. The improvements of 0.9%, 2.0%, and 0.4% in SR, OSR, and SPL respectively demonstrate that instruction modulation through FiLM is crucial for conditioning queries on navigation goals. This validates that affine transformation of queries based on instruction embeddings enables more effective extraction of task-relevant information from the memory.

5. Limitations and Conclusion

In this work, we propose OctMem-Agent, a memory-augmented vision-language navigation framework for aerial object-goal navigation. First, we introduce an Adaptive Octree Memory Representation that incrementally aggregates RGB-D observations into hierarchical 3D representations, enabling effective spatial indexing and scalable memory management across large-scale aerial environments. Second, we develop an Instruction-Guided Memory Query module that extracts task-relevant information through instruction-modulated queries, effectively balancing target localization and frontier exploration. Extensive experiments on the UAV-ON benchmark demonstrate significant improvements over existing methods, validating the effectiveness of our memory-augmented approach. However, our adaptive octree memory representation relies on accurate depth estimation or pose and IMU drift, which may degrade in challenging outdoor conditions such as low-light environments, adverse weather, or textureless regions, leading to erroneous depth estimates. Future work will explore robust depth fusion techniques and multi-sensor integration to improve reliability in diverse environmental conditions.

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